Per-block threshold Brier scores for the time-block forecast

Per-block and per-dataset breakdowns, and why the rule was demoted

Score report

4 August 2026

Abstract

This is the companion to `tex/timeblock_brier_fullseries`. It scores the *same fits*, on the same ten datasets, five blocks and three methods, and differs in exactly one respect: the Brier thresholds are the validation block's own quantiles rather than the dataset's full series. Because the two reports share a cache, the comparison between them is controlled — no difference between them can be attributed to different fits, samples, or priors. That comparison is the point of this report, and it is stronger than the look-ahead objection that originally motivated the demotion. The per-block rule does not shrink the AR(2) advantage; it *inflates the noise around it*. At setting 5, q_{95}, its estimated gap is -18.1×10^{-3}, more than twice the full-series estimate of -8.9×10^{-3}, and yet it reaches $p = 0.110$ where the full-series rule reaches $p = 0.002$. Per-dataset agreement falls from 10/10 to 7/10. The mechanism is measured directly in §**??**: the rule holds the realised exceedance rate at exactly $1-p$ in all fifty (dataset, block) cells — standard deviation zero — so the threshold is a random statistic of the very window being scored, and the per-block randomness it injects survives the pairing. A further symptom is that this rule's estimated gap is nearly *constant* in p, where the full-series gap decays with the base rate as it should.

1 Scope and design

Identical to the sibling report: the Experiment A campaign from `simstudy/output/results/scores{5,7}_hn.RD` seams at $T_o = 50, 80, 110, 140, 170$, $H = 15$, three methods (i.i.d., AR(1), AR(2)), ten datasets, five blocks, $\lambda \sim \mathrm{HN}(0, 20)$, `hn_prior = TRUE`, no missing cells and hence $n = 10$ with no clean-n selection. Settings 5 (near-unit-root AR(2), $\phi = (0.15, 0.80)$) and 7 (ARFIMA$(0, 0.45, 0)$, Hurst 0.95).

The one difference: every table below reads `brier.lead.blockq` instead of `brier.lead.` `scores.R` computes both arrays in the same pass, from the same predictive draws, so the two reports are two scorings of one campaign rather than two campaigns.

Thresholds q_{90} and q_{95} are co-primary; the eleven-point grid is in the appendix. Brier only — CRPS is invariant to the threshold rule and so has nothing to say about the distinction this report exists to draw. The retired campaign's CRPS results are in `tex/timeblock_expABC_legacy`.

2 The per-block threshold rule

For each block the thresholds are taken from the held-out window itself,

$$\tau_p^{\mathrm{blk}}(d, s, b) \;=\; \mathrm{quantile}\big(y_{\mathrm{val}}(d, s, b),\, p\big),$$

recomputed for every (dataset, setting, block) and shared across methods and leads within that block. It was the rule in force for every Brier table written before 2026-07-31, and it is retained in the caches as `brier.lead.blockq`, tagged so that `brier_threshold_basis` distinguishes it and the merge gate refuses to average the two definitions together.

Two objections were raised against it. The first is that it is look-ahead: the threshold is a statistic of the window being forecast, so the scoring rule is not implementable in a real forecast setting. That objection is sound but qualitative. The second is that it pins the exceedance base rate, so a level error cannot move it. §?? measures the pinning, and §?? measures what it costs.

3 Per-block breakdown

Table 1: Experiment A, per-block threshold rule: mean Brier score by block, leads 1–15 and the remaining axis pooled. Levels in native units; the AR(2)−i.i.d. gap in units of 10^{-3} (negative favours AR(2)). Co-primary thresholds q_{90} and q_{95} side by side.

		q_{90}				q_{95}			
setting	block	i.i.d.	AR(1)	AR(2)	gap	i.i.d.	AR(1)	AR(2)	gap
5	1	0.1005	0.1010	0.0917	−8.8	0.0551	0.0562	0.0489	−6.3
	2	0.1341	0.1616	0.1186	−15.5	0.0835	0.1098	0.0687	−14.8
	3	0.1262	0.1303	0.1150	−11.2	0.0789	0.0841	0.0675	−11.4
	4	0.1394	0.1287	0.1065	−32.9	0.0921	0.0830	0.0611	−30.9
	5	0.1742	0.1644	0.1452	−29.0	0.1221	0.1157	0.0951	−27.0
7	1	0.1078	0.1092	0.0997	−8.2	0.0638	0.0655	0.0572	−6.6
	2	0.1023	0.1089	0.1163	+14.0	0.0564	0.0640	0.0704	+14.0
	3	0.1020	0.1011	0.0967	−5.3	0.0544	0.0555	0.0523	−2.1
	4	0.0961	0.1038	0.0980	+1.9	0.0516	0.0590	0.0536	+2.0
	5	0.0959	0.0977	0.1009	+5.0	0.0515	0.0539	0.0568	+5.3

The levels are the tell. Under this rule setting 5's i.i.d. scores at q_{95} run from 0.0551 to 0.1221 across the five blocks — a factor of 2.2 between the easiest and hardest window. Under the full-series rule the same five blocks run from 0.0154 to 0.1060, a factor of 6.9: three times the dynamic range, on identical forecasts. The rule compresses the blocks toward each other because it gives each block a threshold matched to its own turbulence: a quiet window is no longer scored as quiet. The block that most disagrees between the two rules is block 3, the quietest one, whose i.i.d. level rises from 0.0154 to 0.0789.

AR(2) still leads i.i.d. in all five blocks at setting 5, and in two of five at setting 7 — one fewer than the full-series rule reports.

4 Per-dataset breakdown

At setting 5 AR(2) wins 7/10 datasets at both thresholds, against 10/10 (q_{90}) and 9/10 (q_{95}) under the full-series rule — three datasets change side purely from the threshold definition, on identical forecasts. This is the per-dataset face of the variance inflation quantified next.

Table 2: Experiment A, per-block threshold rule: mean Brier score by data set, leads 1–15 and the remaining axis pooled. Levels in native units; the AR(2)−i.i.d. gap in units of 10^{-3} (negative favours AR(2)). Co-primary thresholds q_{90} and q_{95} side by side.

		q_{90}				q_{95}			
setting	data set	i.i.d.	AR(1)	AR(2)	gap	i.i.d.	AR(1)	AR(2)	gap
5	1	0.1067	0.1173	0.0874	−19.3	0.0621	0.0736	0.0472	−14.9
	2	0.1029	0.1484	0.1183	+15.5	0.0557	0.0999	0.0675	+11.9
	3	0.2418	0.2032	0.1054	−136.4	0.1824	0.1476	0.0582	−124.1
	4	0.0967	0.0988	0.1061	+9.4	0.0512	0.0536	0.0582	+7.0
	5	0.1467	0.1424	0.1183	−28.4	0.0989	0.0937	0.0710	−27.9
	6	0.1128	0.1126	0.1099	−2.8	0.0668	0.0673	0.0641	−2.7
	7	0.1239	0.1260	0.1137	−10.2	0.0752	0.0777	0.0662	−9.0
	8	0.1401	0.1399	0.1282	−11.9	0.0963	0.0968	0.0799	−16.4
	9	0.1129	0.1122	0.1508	+37.9	0.0660	0.0658	0.1036	+37.6
	10	0.1643	0.1715	0.1156	−48.7	0.1088	0.1219	0.0666	−42.2
7	1	0.0958	0.0946	0.0941	−1.6	0.0490	0.0486	0.0484	−0.6
	2	0.1118	0.1161	0.1138	+2.0	0.0637	0.0672	0.0657	+2.0
	3	0.1287	0.1432	0.1174	−11.2	0.0836	0.1017	0.0745	−9.1
	4	0.1023	0.1044	0.1035	+1.2	0.0564	0.0581	0.0572	+0.8
	5	0.0993	0.1116	0.1021	+2.8	0.0542	0.0658	0.0588	+4.6
	6	0.1019	0.0985	0.0909	−11.0	0.0520	0.0541	0.0482	−3.8
	7	0.0905	0.0921	0.1006	+10.0	0.0482	0.0500	0.0589	+10.8
	8	0.0929	0.0926	0.0921	−0.8	0.0484	0.0483	0.0481	−0.2
	9	0.0929	0.0930	0.0927	−0.1	0.0491	0.0495	0.0493	+0.2
	10	0.0925	0.0952	0.1159	+23.4	0.0508	0.0524	0.0713	+20.5

Table 3: Experiment A, per-block threshold rule: paired one-sided tests (H_1: the first method of the contrast scores lower), data-set unit $n = 10$ — the lead window is averaged first, then the five blocks. Gaps in units of 10^{-3}; $p < 0.05$ in bold.

			q_{90}			q_{95}		
setting	contrast	leads	gap	t p	W p	gap	t p	W p
5	AR(1)−i.i.d.	1–5	−11.6	0.124	0.154	−11.2	0.118	0.238
	AR(1)−i.i.d.	1–15	+2.3	0.638	0.762	+3.5	0.706	0.889
	AR(2)−i.i.d.	1–5	−23.6	0.094	0.077	−20.6	0.107	0.131
	AR(2)−i.i.d.	1–15	−19.5	0.114	0.111	−18.1	0.110	0.093
	AR(2)−AR(1)	1–5	−12.0	0.099	0.063	−9.4	0.134	0.077
	AR(2)−AR(1)	1–15	−21.8	**0.046**	**0.042**	−21.5	**0.039**	**0.042**
7	AR(1)−i.i.d.	1–5	−2.1	0.229	0.154	−0.5	0.376	0.380
	AR(1)−i.i.d.	1–15	+3.3	0.947	0.949	+4.0	0.968	0.993
	AR(2)−i.i.d.	1–5	−3.0	0.259	0.131	−1.9	0.305	0.207
	AR(2)−i.i.d.	1–15	+1.5	0.675	0.658	+2.5	0.822	0.821
	AR(2)−AR(1)	1–5	−0.9	0.378	0.207	−1.4	0.309	0.131
	AR(2)−AR(1)	1–15	−1.8	0.321	0.131	−1.5	0.347	0.154

5 Paired tests

Same construction as the sibling report: one-sided, dataset unit, $n = 10$, lead window averaged before blocks. Only one contrast attains significance anywhere — AR(2)−AR(1) over leads 1–15, at both thresholds, marginally ($t\,p = 0.046$ and 0.039). The AR(2)−i.i.d. contrast, which the full-series rule resolves at $p \leq 0.004$ everywhere, does not separate at all here. Setting 7 is null throughout, as it is under the other rule.

6 The two rules on identical fits

Table 4: The two threshold rules on *identical fits*: paired one-sided tests at the data-set unit ($n = 10$), leads 1–15. The per-block rule does not shrink the gap — at setting 5, q_{95} it is *twice* the full-series gap — yet it reaches no significance, because the threshold is a statistic of the very window being scored and so injects a per-block random component that breaks the pairing. Gaps in units of 10^{-3}; $p < 0.05$ in bold.

			full-series		per-block	
setting	contrast	p	gap	$t\ p$	gap	$t\ p$
5	AR(1)−i.i.d.	0.90	−1.9	0.365	+2.3	0.638
	AR(1)−i.i.d.	0.95	+1.6	0.737	+3.5	0.706
	AR(2)−AR(1)	0.90	−23.9	<0.001	−21.8	**0.046**
	AR(2)−AR(1)	0.95	−10.5	**0.001**	−21.5	**0.039**
	AR(2)−i.i.d.	0.90	−25.8	**0.003**	−19.5	0.114
	AR(2)−i.i.d.	0.95	−8.9	**0.002**	−18.1	0.110
7	AR(1)−i.i.d.	0.90	+1.9	0.764	+3.3	0.947
	AR(1)−i.i.d.	0.95	+2.6	0.912	+4.0	0.968
	AR(2)−AR(1)	0.90	−2.1	0.250	−1.8	0.321
	AR(2)−AR(1)	0.95	−1.3	0.269	−1.5	0.347
	AR(2)−i.i.d.	0.90	−0.1	0.475	+1.5	0.675
	AR(2)−i.i.d.	0.95	+1.3	0.836	+2.5	0.822

This is the controlled comparison the retired campaign could never make, because its Brier tables and its successors came from different fits. Here the fits, the datasets, the blocks and the predictive draws are the same objects in memory; only the threshold differs.

The rule does not shrink the signal. At setting 5 the per-block rule's point estimates are *larger* in magnitude than the full-series rule's at q_{95} — −18.1 against −8.9 (in 10^{-3}) for AR(2)−i.i.d., and −21.5 against −10.5 for AR(2)−AR(1). If the demotion cost only sensitivity, this is not what one would see.

It inflates the noise. Despite the larger gaps, the per-block rule cannot separate: $p = 0.110$ against 0.002 for AR(2)−i.i.d. at q_{95}. A larger estimate with a much larger p-value means the across-dataset variance has grown faster than the mean. The pairing is what suffers: two methods scored in the same block still share a threshold, but that threshold is itself a draw, and its randomness enters every dataset's difference differently.

It is nearly blind to the threshold. The appendix sweeps both rules across the eleven-point grid. Under the full-series rule setting 5's gap decays smoothly with p, from -25.8 at q_{90} to -0.6 at q_{995}, tracking the base rate. Under the per-block rule it is almost flat — -19.5 at q_{90}, -11.7 at q_{995} — and its p-value barely moves off 0.11 across the entire grid. A scoring rule whose answer does not depend on which tail you ask about is not measuring the tail; it is measuring the fixed $1-p$ of events it has guaranteed itself.

7 Base-rate pinning, measured

The claim that this rule pins the exceedance rate has been asserted before but not shown. It is a statement about the data alone — no fit enters — so it can be checked directly. For each of the fifty (dataset, block) cells the realised exceedance rate of the validation window is computed under both rules.

Table 5: Base-rate pinning, computed from the data alone (no fits are involved). For each of the 50 (data set, block) cells the realised exceedance rate of the validation window is evaluated under each rule. The per-block rule holds it at the nominal $1-p$ with almost no spread; the full-series rule lets it vary with what the block actually did, which is the whole reason a level error is visible under one rule and invisible under the other.

			per-block rule		full-series rule		
setting	p	$1-p$	mean	sd	mean	sd	range
5	0.90	0.100	0.1000	0.0000	0.1257	0.2073	0.000–0.931
	0.95	0.050	0.0500	0.0000	0.0682	0.1333	0.000–0.577
7	0.90	0.100	0.1000	0.0000	0.1111	0.1001	0.000–0.409
	0.95	0.050	0.0500	0.0000	0.0574	0.0669	0.000–0.291

The pinning is exact. Under the per-block rule every one of the fifty cells realises a rate of precisely $1-p$: mean 0.1000 at q_{90} and 0.0500 at q_{95}, with standard deviation *zero* and identical minimum and maximum. Under the full-series rule the same cells range from 0 to 0.9306 at setting 5, q_{90}.

This is the mechanism behind both preceding sections. A window that ran far above its training level cannot register as unusual, because its threshold rose with it; a window in which nothing happened is handed exactly as many "exceedances" as a violent one. The base rate carries no information, so the only channel left to the score is the shape of the predictive around a moving target — which is both a weaker signal and a noisier one.

8 Provenance

Caches, scripts and gates as in the sibling report: `simstudy/output/results/scores{5,7}_hn.RData`, `brier_threshold_basis = "full_series"` recording the primary rule, `hn_prior = TRUE`; tables from `Rscript expA_breakdown.R --prior=hn --settings=5,7`, which emits both rules' fragments from one load and recomputes the pinning table from `simdata.RData`. No number in this document is transcribed by hand.

The `_hn` suffix is the lambda-prior arm, which every per-arm artifact has carried since 2026-08-05. Both this report and its sibling are the HN arm; the threshold rule is the only thing that differs between them.

Sibling: `tex/timeblock_brier_fullseries`. Frozen record of the retired campaign, settings 4 and 6, Experiments B and C, and all CRPS results: `tex/timeblock_expABC_legacy`.

A The full threshold grid

Table 6: Experiment A, per-block threshold rule: the whole threshold grid, leads 1–15. Gap = AR(2)−i.i.d. in units of 10^{-3}, data-set unit $n = 10$. "won" counts the data sets (of 10) and blocks (of 5) in which AR(2) scores below i.i.d.

setting	p	i.i.d.	AR(1)	AR(2)	gap	t p	W p	ds won	blk won
	0.900	0.1349	0.1372	0.1154	−19.5	0.114	0.111	7/10	5/5
	0.910	0.1260	0.1286	0.1068	−19.2	0.113	0.111	7/10	5/5
	0.920	0.1164	0.1192	0.0973	−19.1	0.112	0.111	7/10	5/5
	0.930	0.1069	0.1099	0.0880	−18.9	0.111	0.093	7/10	5/5
	0.940	0.0969	0.1001	0.0783	−18.6	0.110	0.093	7/10	5/5
5	0.950	0.0863	0.0898	0.0683	−18.1	0.110	0.093	7/10	5/5
	0.960	0.0751	0.0788	0.0578	−17.4	0.110	0.093	7/10	5/5
	0.970	0.0631	0.0668	0.0463	−16.8	0.104	0.093	7/10	5/5
	0.980	0.0506	0.0544	0.0349	−15.7	0.102	0.111	7/10	5/5
	0.990	0.0358	0.0394	0.0220	−13.8	0.101	0.093	7/10	5/5
	0.995	0.0264	0.0301	0.0148	−11.7	0.104	0.077	7/10	5/5
	0.900	0.1008	0.1041	0.1023	+1.5	0.675	0.658	5/10	2/5
	0.910	0.0924	0.0959	0.0940	+1.6	0.702	0.730	4/10	2/5
	0.920	0.0834	0.0871	0.0853	+1.9	0.740	0.793	4/10	2/5
	0.930	0.0746	0.0785	0.0768	+2.1	0.768	0.793	4/10	2/5
	0.940	0.0652	0.0692	0.0676	+2.4	0.799	0.793	4/10	2/5
7	0.950	0.0555	0.0596	0.0580	+2.5	0.822	0.821	4/10	2/5
	0.960	0.0461	0.0501	0.0487	+2.6	0.840	0.846	4/10	2/5
	0.970	0.0360	0.0400	0.0386	+2.6	0.860	0.869	4/10	2/5
	0.980	0.0258	0.0297	0.0283	+2.5	0.869	0.907	3/10	2/5
	0.990	0.0145	0.0180	0.0168	+2.3	0.885	0.949	3/10	1/5
	0.995	0.0087	0.0119	0.0106	+1.9	0.886	0.967	2/10	1/5

Per-block rule, pooled over leads 1–15 at the dataset unit. Compare the full-series sweep in the sibling report's appendix: the gap here is nearly constant in p and the p-value never leaves the neighbourhood of 0.11, while the dataset and block counts are frozen at 7/10 and 5/5 across all eleven thresholds — the signature of a score whose base rate has been fixed by construction.